<u>Labaratory Work No.8</u>

Topic: *Solve a system of linear algebraic equations using Kramer's method.*

Questions for self-control

<u>1.What is called a system of linear equations?</u>

A system of linear equations is a set of equations where each equation is linear and involves the same variables. The goal is to find values for the variables that satisfy all equations.

<u>2.What is the solution of a system of linear equations?</u>

The solution of a system of linear equations is the set of values for the variables that make all equations true at the same time. The system can have one solution, infinitely many solutions, or no solution.

<u>3.Matrix notation of a system of linear equations?</u>

In matrix notation, a system of linear equations is written as **AX = B**, where **A** is the coefficient matrix, **X** is the column of variables, and **B** is the column of constants. This form helps in solving the system using matrix operations.

<u>4.Why do I need to find the determinant of the coefficient matrix?</u>

The determinant of the coefficient matrix helps to check if the system has a unique solution. If the determinant is not zero, the system has a single solution. If it is zero, the system either has no solution or infinitely many solutions.

<u>Task 5.</u>

5	$\begin{cases} x_1 + x_2 + x_3 = 2 \\ 2x_1 - x_2 - 6x_3 = -1 \\ 3x_1 - 2x_2 = 8 \end{cases}$

The task requires solving a system of linear algebraic equations using the Cramer's method. In order to solve the task, we need to follow some steps.

$$A := \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & -6 \\ 3 & -2 & 0 \end{bmatrix}$$

$$\Delta := |A|$$

$$\Delta = -31$$

$$A1 := \begin{bmatrix} 2 & 1 & 1 \\ -1 & -1 & -6 \\ 8 & -2 & 0 \end{bmatrix}$$

$$A2 := \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & -6 \\ 3 & 8 & 0 \end{bmatrix}$$

$$A3 := \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & -1 \\ 3 & -2 & 8 \end{bmatrix}$$

$$\Delta 1 := |A1|$$

$$\Delta 2 := |A2|$$

$$\Delta 3 := |A3|$$

$$\Delta 1 = -62$$

$$\Delta 2 = 31$$

$$\Delta 3 = -31$$

$$x1 := \frac{\Delta 1}{\Delta}$$

$$x2 := \frac{\Delta 2}{\Delta}$$

$$x3 := \frac{\Delta 3}{\Delta}$$

$$x1 = 2$$

$$x2 = -1$$

$$x3 = 1$$

The first step is to turn linear equations into matrix form. When rewriting the equations in matrix form, we take each coefficient of **x1, x2** and **x3** and add it in matrix A. From the toolbar **"Matrix"** we choose the template for input. After that, the dialog box appears where we need to input the number of column and rows. As we can see in the given task, I have 3 variables and 3 equations, so the matrix is **3x3**.

It is necessary to check if the Determinant of the matrix is not zero because we cannot solve it in Cramer's Matrix method if it is zero. We select a Determinant icon on Matrix toolbar to find it and denote it as Δ from the Symbols sidebar. After the calculations we discover it is –31 so we can continue to solve the task. And we display the value Δ by typing $\Delta=$.

Construct three matrices as follows: The matrix A1 is obtained from the original matrix A by replacing its first column with the column of constant (free) terms from the system of linear equations. These free terms represent the right-hand side values of the equations in the system, while the rest of the matrix remains unchanged.

The matrix A2 is formed by taking matrix A and replacing its second column with the column of constant (free) terms from the system, which corresponds to the right-hand sides of the equations, while keeping the other columns unchanged.

The matrix A3 is derived from matrix A by substituting its third column with the column of constant (free) terms from the system, which represent the right-hand sides of the equations, while keeping the remaining columns unchanged. Next, we calculate the determinants of these matrices $|A1|$, $|A2|$, $|A3|$ by denoting them as $\Delta 1$, $\Delta 2$ and $\Delta 3$ respectively. We enter the received values on the screen by typing $\Delta 1=$, $\Delta 2=$ and $\Delta 3=$. Then, we type the formulas and find a solution to the system using the formulas.

Finally, display the values $x1$, $x2$, $x3$ on the screen by typing $x1=$, $x2=$, $x3=$. The answer is a triple of numbers (-2, -1, 1).

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$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & -6 \\ 3 & -2 & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 8 \end{pmatrix}$$

$$\det(A) = -31$$

$$\Delta_1 = \begin{vmatrix} 2 & 1 & 1 \\ -1 & -1 & -6 \\ 8 & -2 & 0 \end{vmatrix} = 0 + 2 + (-48) + 8 - 24 - 0 = -62$$

$$\Delta_2 = \begin{vmatrix} 1 & 2 & 1 \\ 2 & -1 & -6 \\ 3 & 8 & 0 \end{vmatrix} = 0 + 16 - 36 + 3 + 48 - 0 = 31$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 2 \\ 2 & -1 & -1 \\ 3 & -2 & 8 \end{vmatrix} = -8 - 8 - 3 + 6 - 2 - 16 = -31$$

$$x_1 = \frac{\Delta_1}{\Delta} = \frac{-62}{-31} = 2$$

$$x_2 = \frac{\Delta_2}{\Delta} = \frac{31}{-31} = -1$$

$$x_3 = \frac{\Delta_3}{\Delta} = \frac{-31}{-31} = 1$$

When implementing the task manually it took more steps and calculations but ultimately it gave the same results.

Conclusion:

In this laboratory work, we examined the use of **SMath Studio** for solving systems of linear equations using **Cramer's Rule**. This method relies on calculating determinants of matrices, making it a straightforward yet powerful approach for solving systems where the determinant of the coefficient matrix is nonzero.

By utilizing **SMath Studio's** built-in functions, such as det(A), we efficiently computed the determinant of the coefficient matrix and the determinants of

modified matrices for each variable. The solutions were then obtained using the formula for each unknown.

The application proved to be an intuitive and effective tool for performing determinant-based calculations, allowing for precise and systematic problem-solving. Using Cramer's Rule in **SMath Studio** not only reinforced the theoretical understanding of determinants but also demonstrated the practical application of this method in solving linear equation systems.